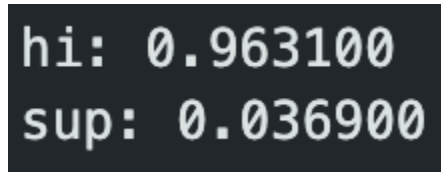


Name: Ethan Jia

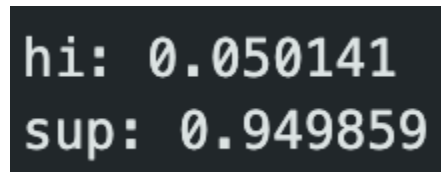
Student ID: ethanjia (2235723)



A screenshot of a serial monitor interface with a black background and white text. It displays two lines of data: 'hi: 0.963100' and 'sup: 0.036900'.

Gesture	Score
hi	0.963100
sup	0.036900

Fig. 1 Screenshot showing hi gesture correctly identified by the Serial Monitor



A screenshot of a serial monitor interface with a black background and white text. It displays two lines of data: 'hi: 0.050141' and 'sup: 0.949859'.

Gesture	Score
hi	0.050141
sup	0.949859

Fig. 2 Screenshot showing sup gesture correctly identified by the Serial Monitor

Brief Explanation:

In the original notebook, the model was compiled using mse as the loss function and mae as the evaluation metric. These settings are more appropriate for regression tasks, but this lab is a multi-class gesture classification task. Since the model uses a softmax output layer and the gesture labels are one-hot encoded, I changed the loss function to categorical_crossentropy and changed the metric to accuracy.

I also updated the downstream plotting code so that it uses accuracy and val_accuracy instead of mae and val_mae. This made the training history keys, plots, metrics, and evaluation results consistent with the corrected classification model. After making these changes, I retrained the model, converted it to a TensorFlow Lite model, and exported the updated model as model.h for Arduino deployment.

Important Changes:

To keep the notebook consistent after fixing the training setup, I updated the metric used in the plotting section from mae to accuracy. This also changed the validation metric from val_mae to val_accuracy, so the training and validation curves now match the corrected classification model. I also reran the notebook after making the changes to make sure the history keys, evaluation output, TensorFlow Lite conversion, and exported model.h were all generated from the updated model. The final notebook output showed accuracy and val_accuracy in the training history, confirming that the corrected metric was used throughout the workflow.